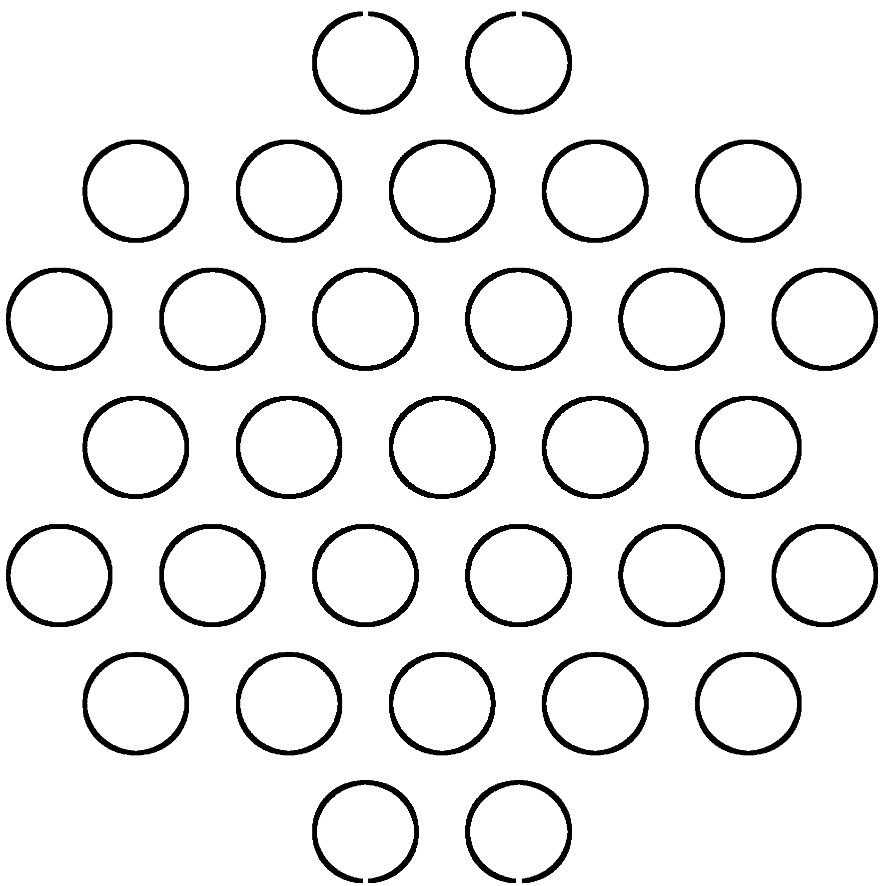
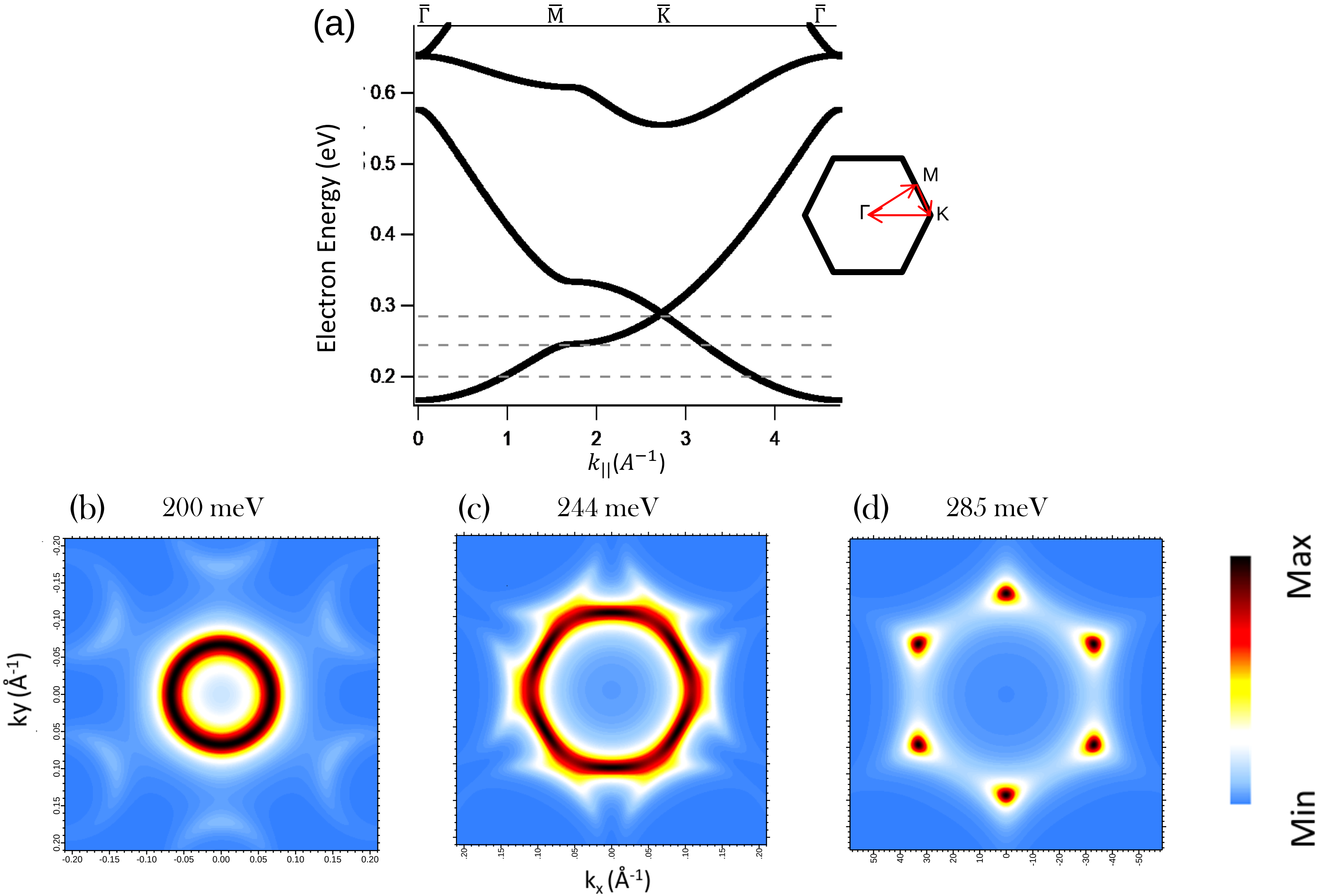


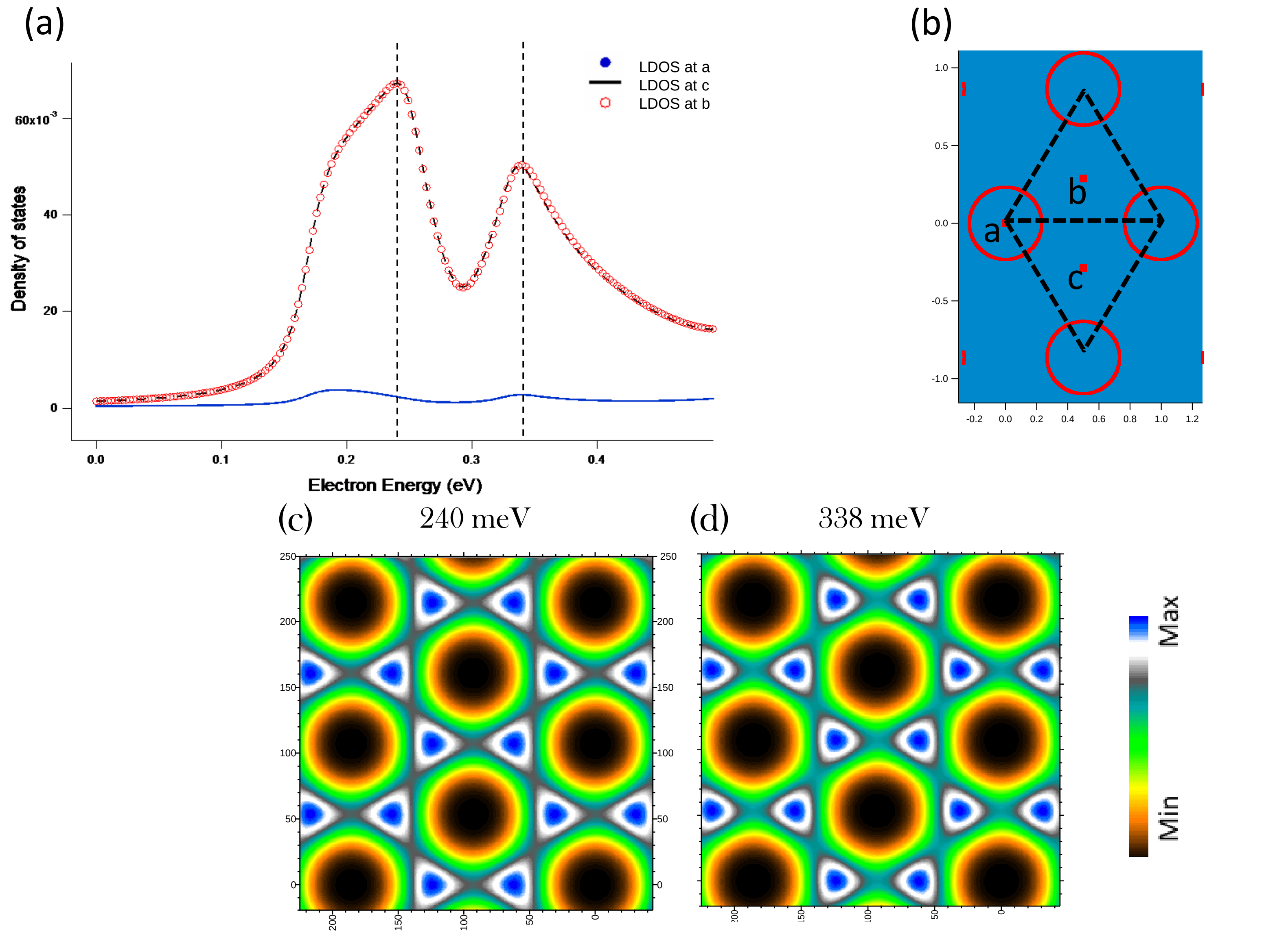
Final data for hexagonal lattce



Fig(1): Band structure and CES of hexagonal metallic lattice

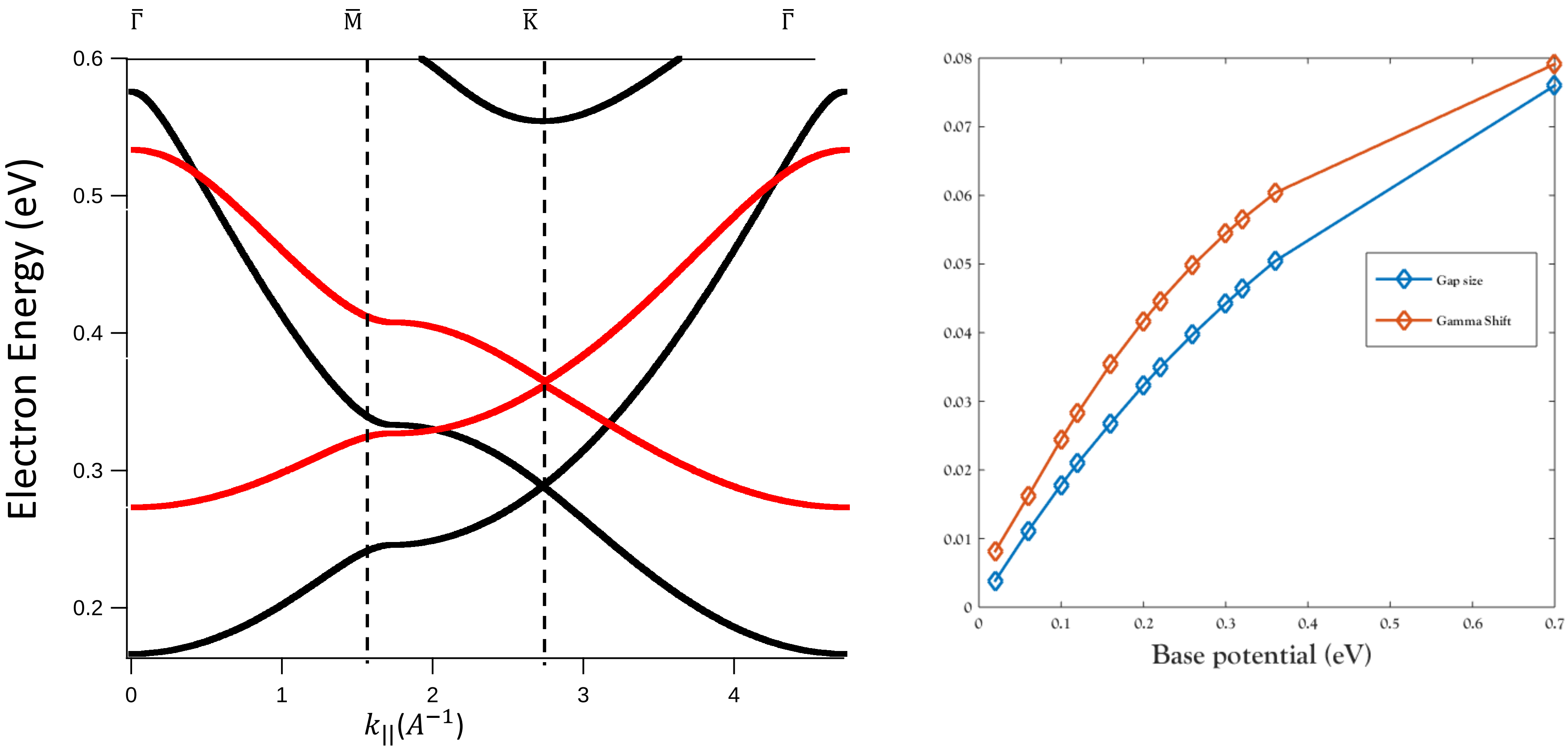
(a) Calculated band structure using EPWE along the $\overline{\Gamma MK\Gamma}$ directions of first brillouin zone arrows in black hexagon.

(b,c,d) Simulated Constant energy surfaces taken at the black dashed lines in (a), i.e. at 200, 244 and 285 mille electron-volt respectively.



Fig(2): LDOS of hexagonal lattice with circular scatters – (a) The local density of states (LDOS) calculated by EPWE for three points in the unit cell, as shown in (b). (c,d) The 2D-LDOS taken at 240 meV and 338 meV.

(a)



Fig(3): Effect of Periodic Potential on the hexagonal Electronic Structure
(a) Calculated band structures using EPWE for two hexagonal metallic super lattices of different potential barrier V_1, along the $\overline{\Gamma M K \Gamma}$ directions. All calculation parameters were the same in the two cases, except that V_1 is changed from 700 meV (black band) to 30 eV (red band).
(b) The Γ-point energy (red) and M-gap size (blue) as a function of the base potential barrier V_1